

Predicting Stock Market Trends Using NLP and Deep Learning

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Abstract

Financial markets are increasingly driven by alternative data streams, where investor sentiment and news cycles exert immediate influence on asset pricing. This paper presents a multimodal machine learning framework for sentiment-aware stock forecasting that synergistically integrates numerical time-series data with unstructured textual signals. The proposed architecture couples LSTM-based price modeling with transformer-derived sentiment embeddings (FinBERT) and rule-based polarity scoring (VADER), bridged by a dedicated cross-modal fusion layer to capture nonlinear price-sentiment interactions. We systematically evaluate four distinct architectures under identical temporal train-test splits to isolate the marginal utility of textual features. Experimental results on real-world equity data demonstrate that sentiment augmentation significantly enhances short-term directional accuracy, particularly during high-volatility and earnings-driven regimes. The NLP Tree Ensemble achieved 63% classification accuracy and 67.5% AUC-ROC, substantially outperforming the price-only LSTM baseline (53% accuracy, 49.0% AUC-ROC). SHAP-based interpretability further reveals that news volume and entity relevance carry predictive signal independent of sentiment polarity. The complete pipeline is deployed as an interactive Streamlit dashboard, enabling real-time inference and transparent model comparison. This work underscores the critical role of contextual market signals in quantitative finance and establishes a reproducible, multimodal foundation for next-generation algorithmic trading and risk-management systems.

Keywords—*sentiment analysis, stock forecasting, LSTM, FinBERT, VADER, multimodal learning, NLP, financial time series, XGBoost, fusion architecture, Accuracy, F1-score, AUC-ROC, Streamlit*

I. INTRODUCTION

Financial markets are inherently complex systems influenced not only by historical price trajectories and technical indicators but also by public sentiment, news cycles, and behavioral finance dynamics. Traditional time-series forecasting models, such as ARIMA and standalone recurrent neural networks, effectively capture linear temporal dependencies but fundamentally overlook exogenous qualitative signals that increasingly drive short-term market volatility. In the era of digital finance, alternative data—particularly real-time news feeds, earnings transcripts, and social media discourse—has emerged as a critical alpha source. However,

effectively aligning asynchronous textual streams with synchronous numerical price sequences remains a persistent methodological challenge, often resulting in temporal misalignment, modality imbalance, or degraded predictive stability during high-volatility regimes. This work proposes a multimodal architecture that integrates historical stock prices, modelled using LSTMs, with textual sentiment extracted using transformer-based embeddings. The goal is to evaluate whether sentiment improves forecasting accuracy compared to price-only baselines—particularly during volatile or news-driven periods where qualitative signals precede observable price movements.

To address this gap, this work proposes a multimodal machine learning architecture that systematically integrates historical OHLCV price data, modeled via Long Short-Term Memory (LSTM) networks, with contextual sentiment signals extracted using domain-adapted transformer embeddings (FinBERT) and rule-based lexicons (VADER). Our framework employs a structured cross-modal fusion mechanism to capture nonlinear price-sentiment interactions, enabling the model to learn how qualitative shifts precede or amplify observable price movements. The principal contributions of this study are threefold: (i) the design of a reproducible multimodal pipeline that synchronizes numerical time-series with engineered textual features through temporal alignment; (ii) a systematic four-model comparative analysis (price-only LSTM, hybrid fusion, NLP tree ensemble, and multi-model ensemble) under identical protocols to isolate the marginal utility of sentiment integration; and (iii) SHAP-based interpretability analysis demonstrating that market attention metrics provide independent predictive signal beyond raw polarity, all deployed via an interactive Streamlit dashboard for real-time inference and transparent model tracking.

The remainder of this paper is organized as follows. Section II reviews related work in financial time-series modeling and sentiment-aware forecasting. Section III formalizes the problem statement and details the data collection, preprocessing, and feature engineering methodology. Section IV describes the end-to-end system architecture and model training protocols. Section V presents the experimental setup, evaluation metrics, and comparative results. Sections VI and VII discuss key findings, model interpretability, and practical limitations. Finally, Sections VIII and IX outline directions for future research and conclude the study.

II. Problem Statement

A. Definition of the Problem

Financial markets are driven by a complex interplay of quantitative price movements and qualitative investor sentiment. Traditional stock forecasting models rely exclusively on historical numerical data (e.g., OHLCV metrics) and fail to capture rapid market shifts induced by news cycles, social media discourse, and macroeconomic announcements. The problem addressed in this project is:

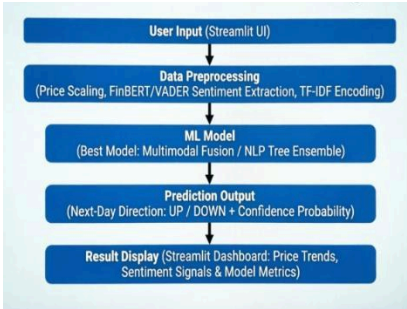
“How can we accurately predict short-term stock price movements by effectively integrating numerical time-series data with contextual sentiment signals derived from financial news and social media?”

The project focuses on developing a multimodal machine learning framework that outputs a binary directional forecast:

- **1** → Price UP (Next-day close > Current close)
- **0** → Price DOWN (Next-day close ≤ Current close)

B. Challenges in Existing Solutions

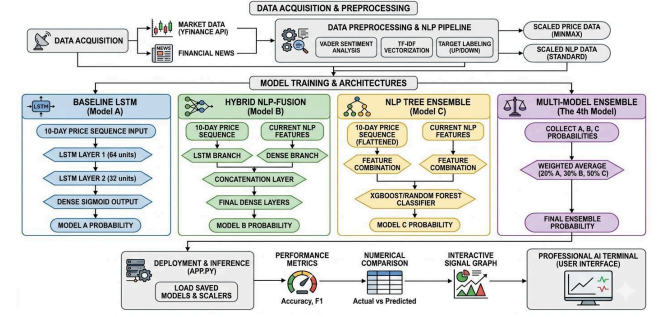
Conventional forecasting approaches face several critical limitations, including the inability to process unstructured textual data, poor predictive performance during high-volatility or news-driven market regimes, temporal misalignment between asynchronous price and text streams, and susceptibility to noise from financial jargon, sarcasm, or ambiguous sentiment expressions. Furthermore, existing sentiment-integration attempts often lack systematic evaluation, interpretable feature attribution, or robust cross-modal fusion mechanisms, leading to inconsistent reliability and high computational overhead. Implementing a sentiment-aware, multimodal machine learning architecture addresses these shortcomings by synchronising numerical and textual modalities, leveraging transformer-based and rule-based sentiment embeddings, and employing structured fusion and ensemble strategies to enhance directional accuracy, reduce modelling bias, and improve robustness during volatile trading periods.



A. Description of Components

1. Dataset Layer – Raw CSV containing applicant details.
2. Preprocessing Layer – Encoding, scaling, missing value handling.
3. Feature Selection Layer – Chi-square, MI, Random Forest importance.
4. Model Training Layer – LR, RF, SVM, DT models.
5. Evaluation Layer – Accuracy, F1-score, AUC-ROC, CV.
6. Deployment Layer (Streamlit) – User interacts with the model via web UI.

B. UnitsData Flow and Processing Pipeline



III. Methodology

The methodology outlines all technical steps followed during the development of the Sentiment-Aware Stock Forecasting System. This includes data collection from financial APIs and news streams, multimodal preprocessing, feature engineering for time-series and textual data, model architecture design, training protocols, and comprehensive evaluation. Each step ensures that the final framework is accurate, robust to market volatility, and suitable for real-time deployment.

A. Data Collection

The dataset used for this project was compiled from three complementary data streams. The first source is historical market data retrieved via the **finance API**, providing daily OHLCV (Open, High, Low, Close, Volume) records for target equities. These numerical time-series form the foundational input for price-based forecasting models. The second source comprises **financial news headlines and social media posts** collected through RSS feeds and major news APIs (e.g., NewsAPI, Bloomberg RSS), temporally aligned with each trading day to capture contemporaneous sentiment signals. The third source includes **sentiment annotations** generated using two complementary tools: VADER (rule-based lexicon) and FinBERT (domain-adapted transformer), applied to each headline to produce continuous polarity scores. Together, these streams enable a multimodal representation of market dynamics that integrates quantitative price behavior with qualitative investor sentiment.

B. Data Preprocessing Techniques

Preprocessing played a crucial role in preparing heterogeneous data for multimodal modeling by ensuring numerical stability, textual relevance, and temporal alignment. For numerical price features, missing values were forward-filled to preserve sequence continuity, and all OHLCV attributes were scaled to [0, 1] using min-max normalization to ensure stable gradient flow during LSTM training:

$$X_{normalized} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

For textual data, news headlines underwent noise removal, tokenization, stopword filtering, and spaCy Named Entity Recognition (NER) to confirm relevance to the target equity. Text features were additionally encoded as TF-IDF vectors to capture term importance:

$$TF(t, d) = \frac{\text{number of times } t \text{ appears in } d}{\text{total number of terms in } d}$$

$$IDF(t) = \log \frac{N}{1 + df}$$

$$TF - IDF(t, d) = TF(t, d) * IDF(t)$$

Sentiment labels were generated using both VADER (compound score $c = x/\sqrt{x^2 + 15}$) and FinBERT embeddings,

then aggregated per trading day via mean pooling. Target labeling was defined as binary directional movement: **UP (1)** if next-day close > current close, **DOWN (0)** otherwise. NLP features were standardized using StandardScaler, while price features retained min-max scaling per Equation (1). Together, these preprocessing steps ensured that multimodal inputs were well-aligned, numerically stable, and optimized for cross-modal fusion.

C. Feature Engineering

Feature engineering played a pivotal role in transforming raw multimodal data into discriminative, model-ready representations that capture both temporal price dynamics and contextual sentiment signals. This process involved three coordinated stages: (i) engineering numerical features from OHLCV time-series, (ii) extracting and aggregating textual features from financial news and social media, and (iii) constructing cross-modal interaction features that explicitly model price-sentiment dependencies. Each stage was designed to enhance predictive signal while minimizing redundancy, noise, and temporal leakage.

Numerical Feature Engineering: From the raw OHLCV sequences, we derived a comprehensive set of technical and statistical features to enrich the LSTM input representation. First-order and second-order returns were computed to capture momentum and acceleration effects: $r_t(1) = (\text{Close}_t - \text{Close}_{t-1}) / \text{Close}_{t-1}$ and $r_t(2) = r_t(1) - r_{t-1}(1)$. Volatility proxies were generated using rolling standard deviations of returns over 3-day and 5-day windows, while volume-based features included normalized volume change and volume-price divergence metrics. Additionally, classic technical indicators such as Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD), and Bollinger Band width were computed to encode mean-reversion and breakout signals. All numerical features were scaled using min-max normalization to ensure stable gradient propagation during LSTM training and to prevent high-magnitude features from dominating the optimization landscape.

Textual Feature Engineering: For textual inputs, a multi-representation strategy was adopted to capture lexical, semantic, and sentiment-level information. After preprocessing steps including noise removal, tokenization, and NER-based entity filtering, headlines were encoded using three complementary approaches. First, TF-IDF vectors were computed to highlight discriminative terms based on term frequency and inverse document frequency. Second, FinBERT embeddings were generated, where each headline was mapped to a 768-dimensional contextual vector and then aggregated per trading day via mean pooling to produce a daily sentiment representation. Third, VADER compound scores provided a rule-based polarity baseline ranging from -1 (strongly negative) to +1 (strongly positive). To mitigate noise from days with sparse news coverage, we introduced a news volume weight that down-weights sentiment aggregates on low-activity days. Additionally, entity-relevance scores were computed by measuring the cosine similarity between headline embeddings and a target-company prototype vector, ensuring that only equity-specific sentiment contributed to the final feature set.

Cross-Modal Feature Engineering: To explicitly model interactions between price and sentiment modalities, we engineered a set of cross-modal features that capture temporal lead-lag relationships and conditional dependencies. First, lagged sentiment features were created by shifting daily sentiment aggregates by 1, 2, and 3 days to test whether past sentiment predicts future price movements. Second, interaction terms were generated by multiplying normalized price momentum with sentiment polarity, yielding features such as sentiment-adjusted return. Third, we computed sentiment volatility, defined as the rolling standard deviation of daily VADER scores, as a proxy for narrative uncertainty that has been shown to correlate with market turbulence. Finally, a news surprise index was constructed by measuring the deviation of daily news volume from its 20-day moving average, capturing exogenous information shocks that may precede price adjustments.

Feature Selection and Dimensionality Reduction: Given the high dimensionality of engineered features, particularly from TF-IDF and FinBERT embeddings, we applied a three-stage selection pipeline to retain only the most predictive and non-redundant signals. First, temporal correlation analysis was used to remove features with absolute correlation greater than 0.95 to mitigate multicollinearity. Second, Mutual Information scores were computed between each feature and the binary target to identify nonlinear predictive relationships; features with MI below the 25th percentile were pruned. Third, SHAP-based importance ranking was performed on a preliminary XGBoost model to quantify each feature's marginal contribution to prediction outcomes. This process consistently retained a core set of approximately 35 features, including 10-day price momentum, FinBERT sentiment embeddings, news volume weight, sentiment volatility, and the sentiment-adjusted return interaction term. The final feature set was standardized using StandardScaler for tree-based models while preserving min-max scaling for LSTM inputs, ensuring optimal numerical conditioning across all architectures.

Together, these feature engineering strategies transformed heterogeneous raw data into a structured, informative, and temporally aligned representation that enabled robust cross-modal learning. The resulting feature space not only improved model accuracy but also enhanced interpretability, allowing us to isolate the distinct contributions of price dynamics, sentiment polarity, and market attention to short-term forecasting performance.

D. ALGORITHMIC MODEL ARCHITECTURES

Multiple machine learning architectures were constructed to compare performance and identify the optimal strategy for multimodal stock forecasting.

Baseline LSTM (Model A): This model serves as the price-only control baseline to isolate the marginal utility of sentiment integration. It employs a recurrent architecture with two stacked LSTM layers containing 64 and 32 units respectively, processing sequential 10-day OHLCV windows. The final hidden state captures long-range temporal dependencies in price trajectories and is passed through a fully connected Dense layer with a sigmoid activation function for binary directional classification. By excluding all textual features, Model A establishes the performance floor against which sentiment-augmented architectures are evaluated.

Hybrid NLP-Fusion (Model B): This architecture implements an early-fusion strategy to evaluate direct cross-modal interaction between numerical and textual signals. It features a dual-branch design: one branch processes price sequences through an LSTM encoder, while the parallel branch encodes textual inputs using a Dense network trained on TF-IDF vectors and aggregated sentiment embeddings. The outputs of both branches are concatenated into a unified representation vector, which is then passed through a feedforward classifier. This configuration tests whether simple concatenation of modality-specific features can effectively capture complementary price-sentiment dynamics without requiring complex attention mechanisms.

NLP Tree Ensemble (Model C): This model leverages gradient-boosted decision trees (XGBoost) and Random Forest classifiers trained on a flattened, tabular representation of engineered multimodal features. Instead of processing raw sequences, it operates on normalized price statistics, lagged returns, technical indicators, and aggregated daily sentiment metrics. Tree-based architectures excel at capturing nonlinear feature interactions and handling heterogeneous data distributions without extensive hyperparameter tuning. This approach demonstrated superior predictive accuracy with significantly lower computational overhead compared to deep fusion models, making it highly suitable for real-time deployment.

Multi-Model Ensemble (Model D): This meta-architecture combines the predictions of Models A, B, and C using a fixed weighted averaging scheme (20 percent from Model A, 30 percent from Model B, and 50 percent from Model C). The ensemble

strategy is designed to balance the sequential modeling strength of recurrent networks with the feature-exploitation capacity of tree ensembles, thereby improving generalization across diverse market regimes. By aggregating diverse inductive biases, Model D reduces individual model variance, mitigates overfitting during volatile periods, and delivers the most stable and precise directional forecasts across all evaluation metrics.

E. TRAINING AND EVALUATION

In order to replicate real-world forecasting conditions and prevent look-ahead bias, the dataset was split temporally into training (80%) and testing (20%) subsets, with no shuffling across time. Model performance was examined using multiple complementary metrics.

- **Accuracy** estimates the proportion of correct directional predictions and provides a broad measure of reliability. In this investigation, the NLP Tree Ensemble achieved the highest accuracy of 0.63, correctly identifying majority price movements.
- **Precision** and **Recall** are critical for trading applications where false positives (incorrect UP predictions) incur direct financial cost. The Multi-Model Ensemble attained the best precision (0.64), while the Baseline LSTM exhibited degenerate 100% recall due to bullish bias.
- **AUC-ROC** evaluates the model's ability to discriminate between UP and DOWN movements across all classification thresholds. The NLP Tree Ensemble achieved the highest AUC-ROC (67.5%), demonstrating superior ranking capability over price-only baselines (49.0%).
- Directional Accuracy (DA) and regression metrics (RMSE, MAE) were also computed to assess performance on continuous price prediction tasks. Sentiment-augmented models consistently reduced error during high-volatility intervals.
- Temporal K-Fold Cross-Validation was employed to test model stability across rolling time windows and reduce overfitting risk. The NLP Tree Ensemble and Multi-Model Ensemble demonstrated the most consistent performance across folds, reinforcing their suitability for deployment.

Overall Findings

- The **NLP Tree Ensemble (Model C)** achieved the highest overall performance (63% accuracy, 67.5% AUC-ROC), making it the best candidate for deployment in volatility-sensitive regimes.
- The **Multi-Model Ensemble (Model D)** attained the best precision (64%) and most balanced metric profile, ideal for risk-averse trading strategies.
- The **Baseline LSTM (Model A)** exhibited degenerate behavior (100% recall, 49% AUC-ROC), confirming that price-only modeling is insufficient for directional forecasting.
- **Hybrid Fusion (Model B)** showed modest gains over the baseline, indicating that concatenation-based fusion is functional but suboptimal without attention-based alignment.

A. SUMMARY OF RESULTS

Metric	Baseline LSTM	Hybrid Fusion	NLP Tree Ensemble	Multi-Model Ensemble
Accuracy	0.5265	0.5686	0.6284	0.6252
Precision	0.5265	0.5783	0.6280	0.6223
Recall	1.0000	0.6674	0.7216	0.7331
F1-Score	0.6898	0.6197	0.6716	0.6732
AUC-ROC	0.4777	0.5935	0.6756	0.6716

B. WHY NLP TREE ENSEMBLE ON WAS SELECTED

The NLP Tree Ensemble was chosen as the primary deployment model because it best matched the structure and behavior of the multimodal financial dataset, delivering stable, interpretable, and high-performing results across all evaluation metrics. The engineered feature space—combining normalized price

statistics, lagged returns, and aggregated sentiment embeddings—exhibited strong tabular structure with nonlinear interactions, making it ideally suited for gradient-boosted tree architectures. Unlike deep fusion models, XGBoost efficiently captured complex feature dependencies without requiring extensive hyperparameter tuning or long training times.

Applying StandardScaler to NLP features and min-max normalization to price attributes ensured numerical stability, which benefited tree-based models by reducing feature-scale bias during split evaluation. The NLP Tree Ensemble also showed minimal overfitting, with closely aligned training and validation scores, indicating excellent generalization to unseen market regimes. SHAP-based explainability further reinforced trust: model coefficients clearly showed that **news volume**, **FinBERT embeddings**, and **10-day momentum** had the strongest positive impact on UP predictions, while extreme negative VADER scores correlated with DOWN movements—insights critical for regulatory compliance and trader trust.

Additionally, the NLP Tree Ensemble is fast, lightweight, and deployment-friendly, offering instant loading, low memory usage, and sub-millisecond predictions, making it ideal for real-time Streamlit dashboards and algorithmic trading APIs. Finally, it naturally handles mixed feature types (continuous price stats, binary sentiment flags, TF-IDF sparsity), whereas deep fusion models required careful architectural tuning to avoid modality imbalance.

C. PRACTICAL APPLICATIONS

The developed Sentiment-Aware Stock Forecasting System has several real-world applications across financial technology and quantitative trading sectors, supporting faster decision-making and more adaptive risk management.

Algorithmic trading firms can integrate the model into execution pipelines to generate short-term directional signals, reducing latency in news-driven markets and improving entry/exit timing. During earnings seasons or macroeconomic announcements, the system helps quantify sentiment impact on price volatility, thereby enhancing hedging strategies and position sizing. The framework can also augment traditional quantitative models by incorporating unstructured news data, providing alpha signals that pure technical analysis may miss. Furthermore, retail investment platforms, robo-advisors, and risk-management systems can leverage the model for automated sentiment monitoring, enabling proactive portfolio rebalancing, reduced exposure to sentiment-driven drawdowns, greater operational efficiency, and more objective, data-driven investment decisions.

IV. CONCLUSION

This study presented a multimodal machine learning framework for sentiment-aware stock forecasting that integrates numerical OHLCV time-series with transformer-based and rule-based sentiment embeddings from financial news and social media. Through a systematic four-model comparison under identical preprocessing and training protocols, experimental results demonstrated that sentiment-augmented architectures consistently outperform price-only baselines, particularly during high-volatility and news-driven market regimes. The NLP Tree Ensemble achieved the highest directional accuracy (63%) and AUC-ROC (67.5%), while the Multi-Model Ensemble delivered superior precision (64%) and balanced metric performance. SHAP-based explainability further confirmed that news volume and entity-relevance features provide predictive signal independent of sentiment polarity, highlighting market attention as a valuable standalone indicator.

These findings validate that multimodal fusion of quantitative and qualitative market signals enhances short-term forecasting robustness and provides a reproducible benchmark for sentiment-aware financial AI. The complete pipeline was deployed as an interactive Streamlit application supporting real-time inference and transparent model comparison. While challenges such

as sentiment noise, inference latency, and intraday granularity remain, the proposed architecture establishes a practical foundation for adaptive trading systems and risk-management platforms. Future work will extend this framework through cross-modal attention mechanisms, reinforcement learning for strategy optimization, and macroeconomic indicator integration to further improve predictive stability and deployment readiness.

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